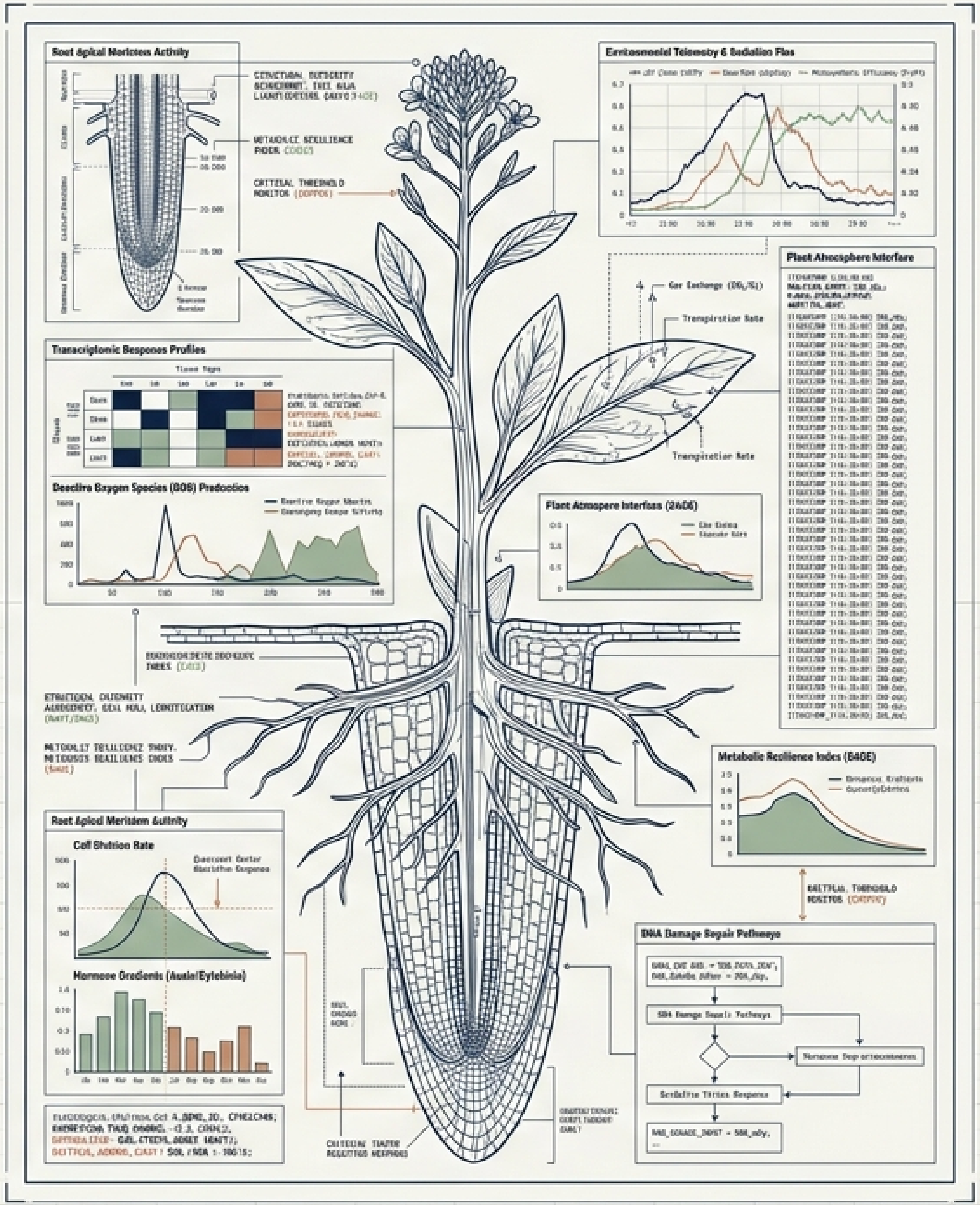
The Kinetic Landscape of Plant Ionizing-Radiation Response

Transcriptomic analysis in spaceflight-relevant environments.

Dr. Richard Barker



Harmonizing A Massive Open-Science Cohort

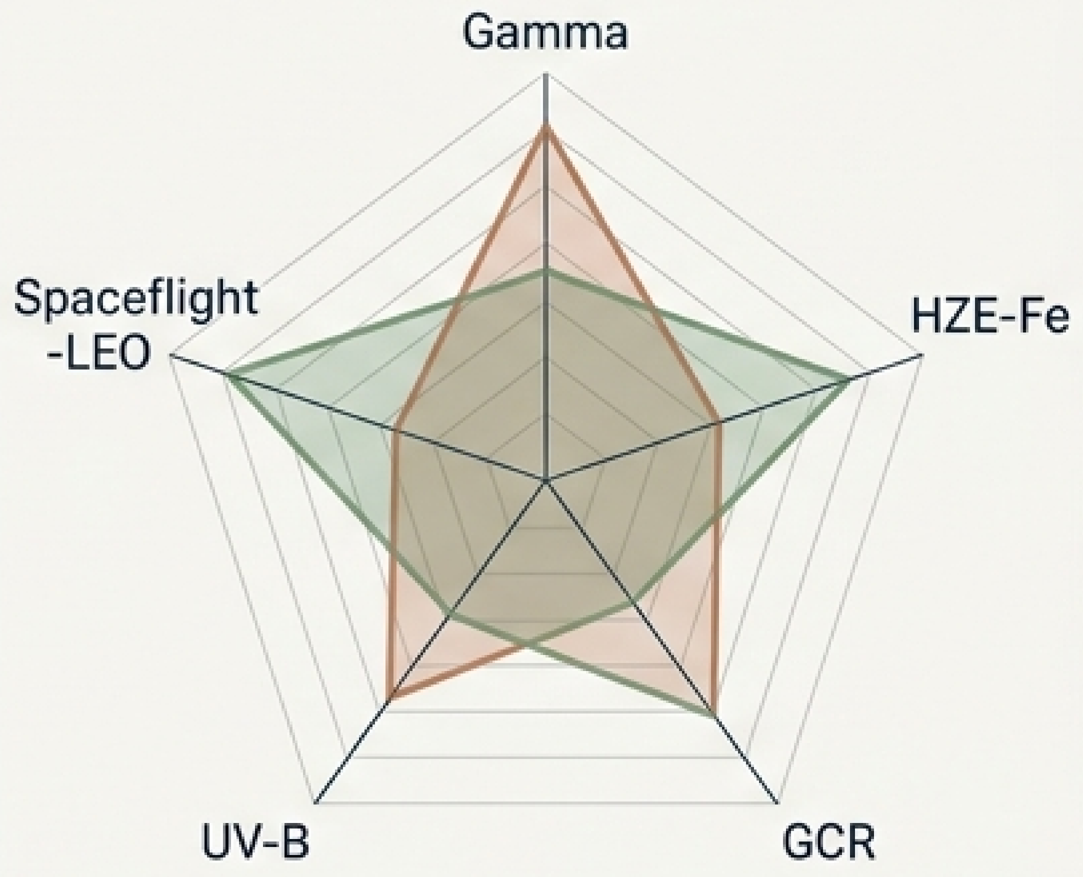
299

Samples

10

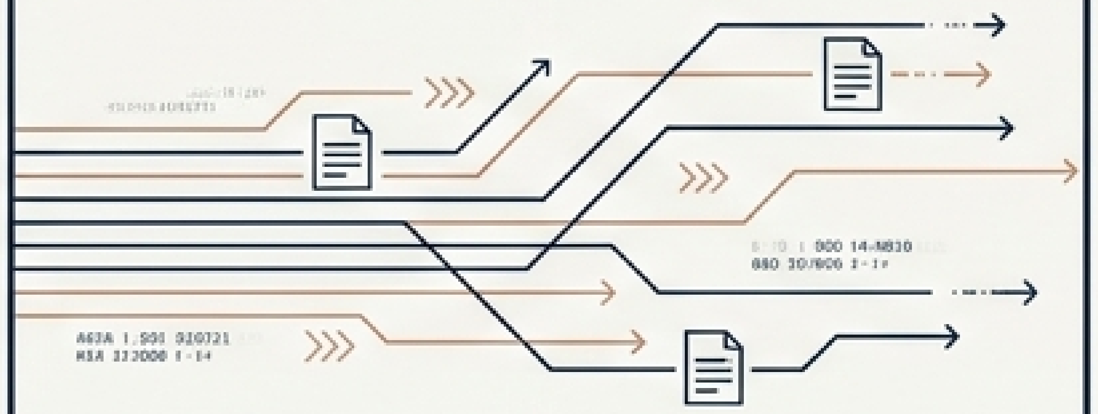
NASA OSDR Studies

5 Radiation Qualities



Data Source

NASA Open Science Data Repository (OSDR)



Bulk transcriptomes processed as raw input for downstream kinetic modeling.

Transitioning from Static Snapshots to Dynamic Continuous Modeling

Legacy Analysis

- Approach: Per-contrast, static analysis.
- Tools: DESeq2 differential expression.
- Scope: Simple knockouts (*sog1-1*, *MYB3R*).
- Status: Preserved in the `legacy/` directory for baseline consistency.

Evolution of the Pipeline

Current Focus

- Approach: Kinetic, cross-study continuous modeling.
- Tools: Gaussian-Process autoencoders & pseudo-cell deconvolution.
- Scope: Dynamic covariates (dose, time, LET) across all 10 studies.

The 5-Step Kinetic Modeling Architecture

1. Aggregation

Merging bulk transcriptomes (n=299)

2. Deconvolution

Generating pseudo-cell-type profiles

3. Modeling

Gaussian-Process autoencoder with dynamic covariates

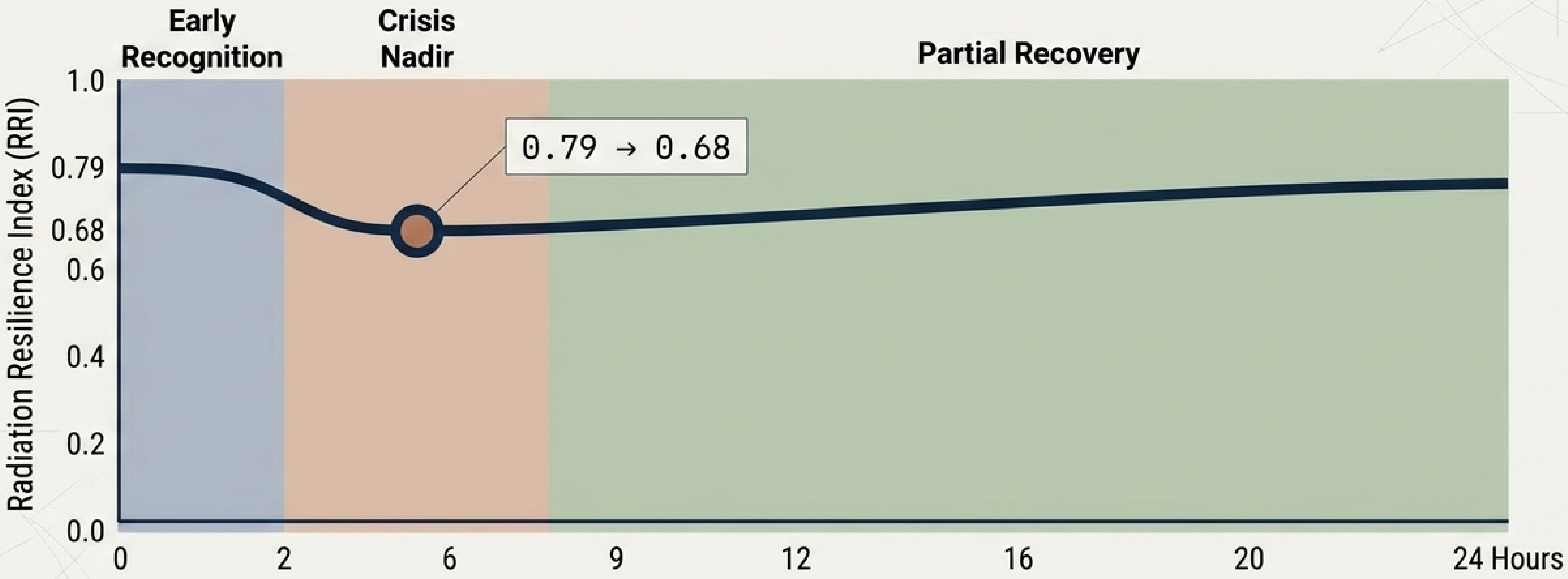
4. Reconstruction

Inter-tissue signaling mapping via `PlantCellChat`

5. Visualization

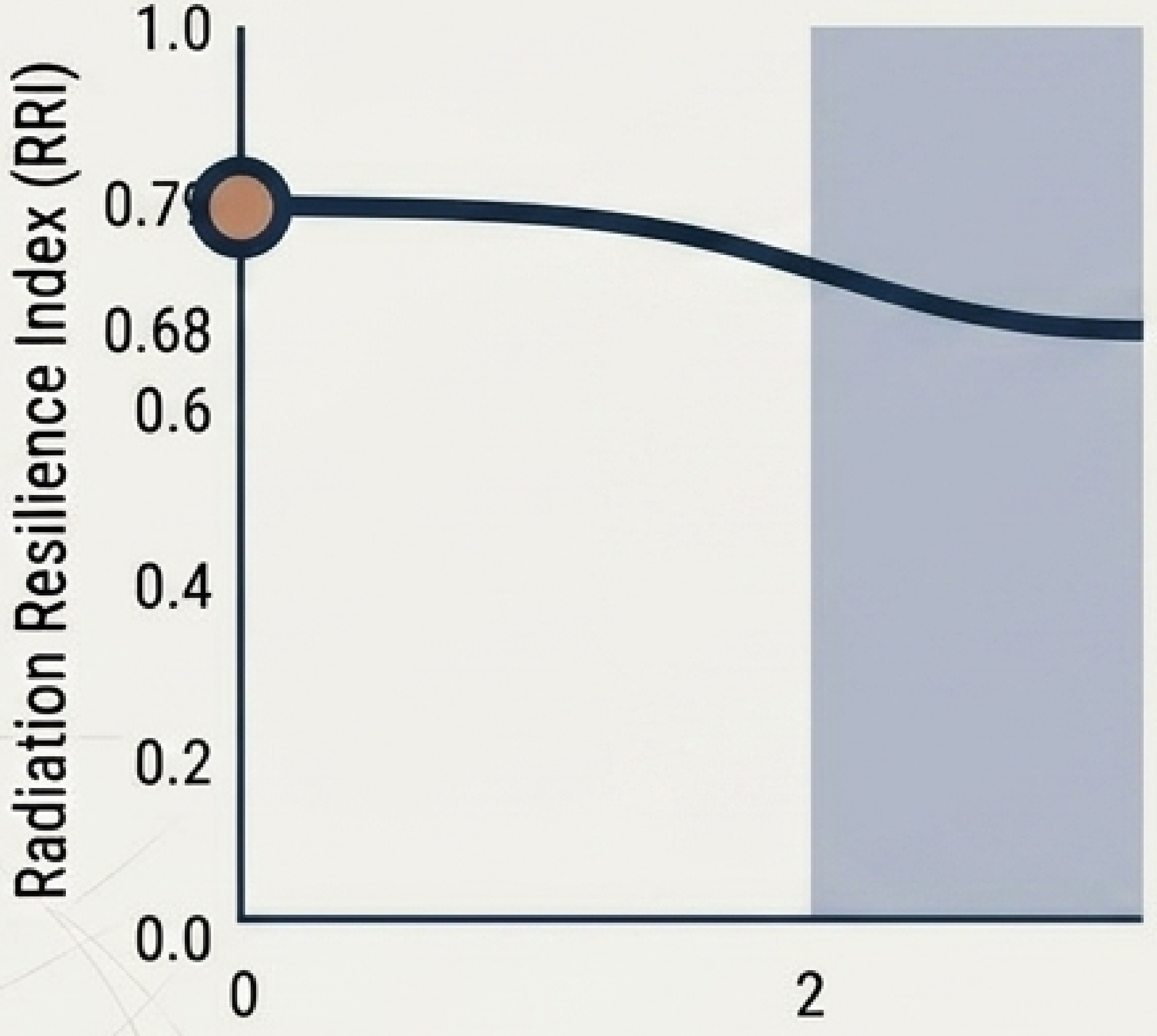
Kinetic modules via `WGCNA` & spatial mapping via `ggPlantmap`

The 3-Phase Kinetic Response to Spaceflight Radiation

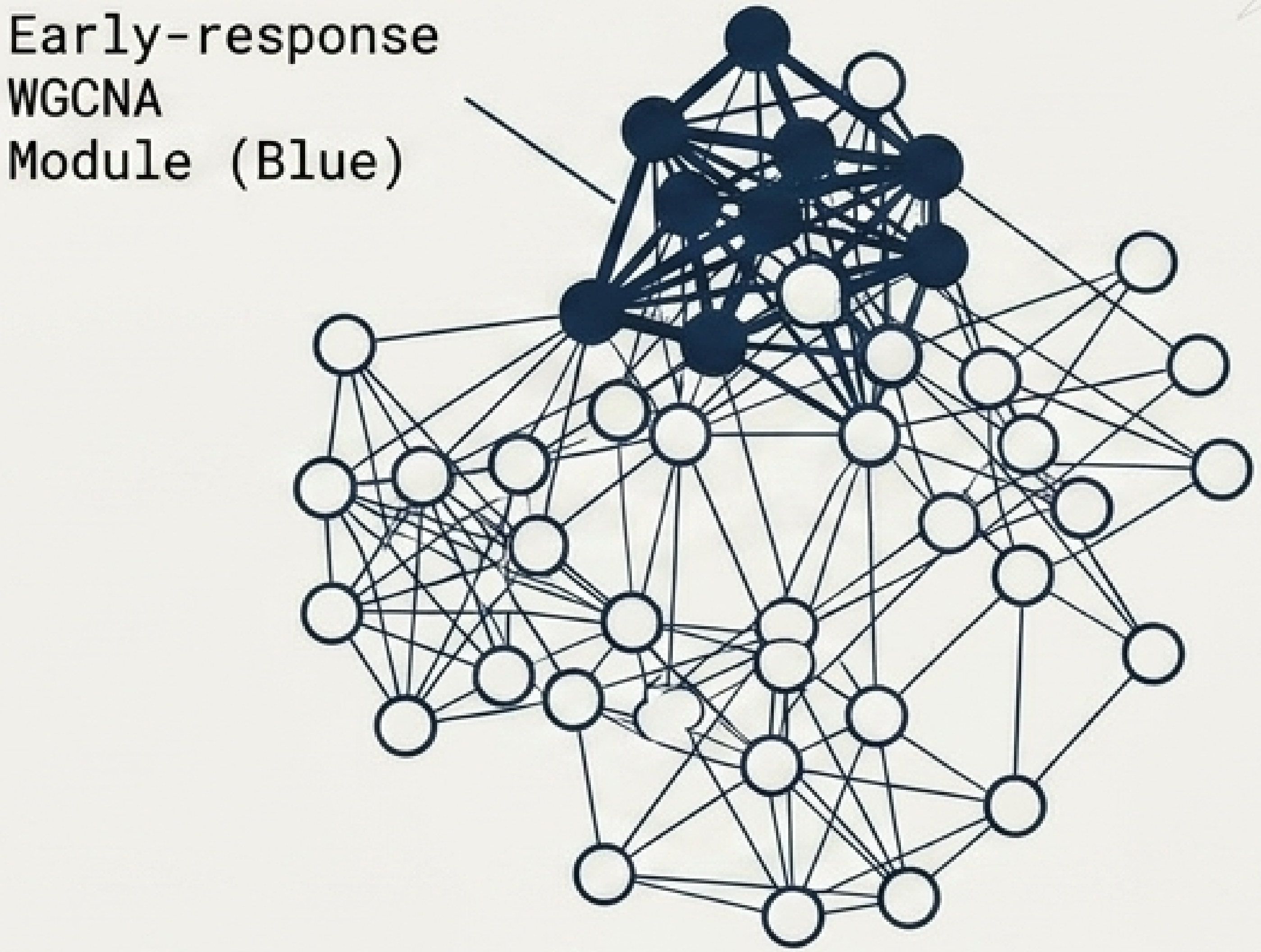


The Radiation Resilience Index (RRI) reveals that radiation stress is highly temporal, requiring dynamic rather than static analysis.

Phase 1: Early Recognition & The Initial Stress Signal



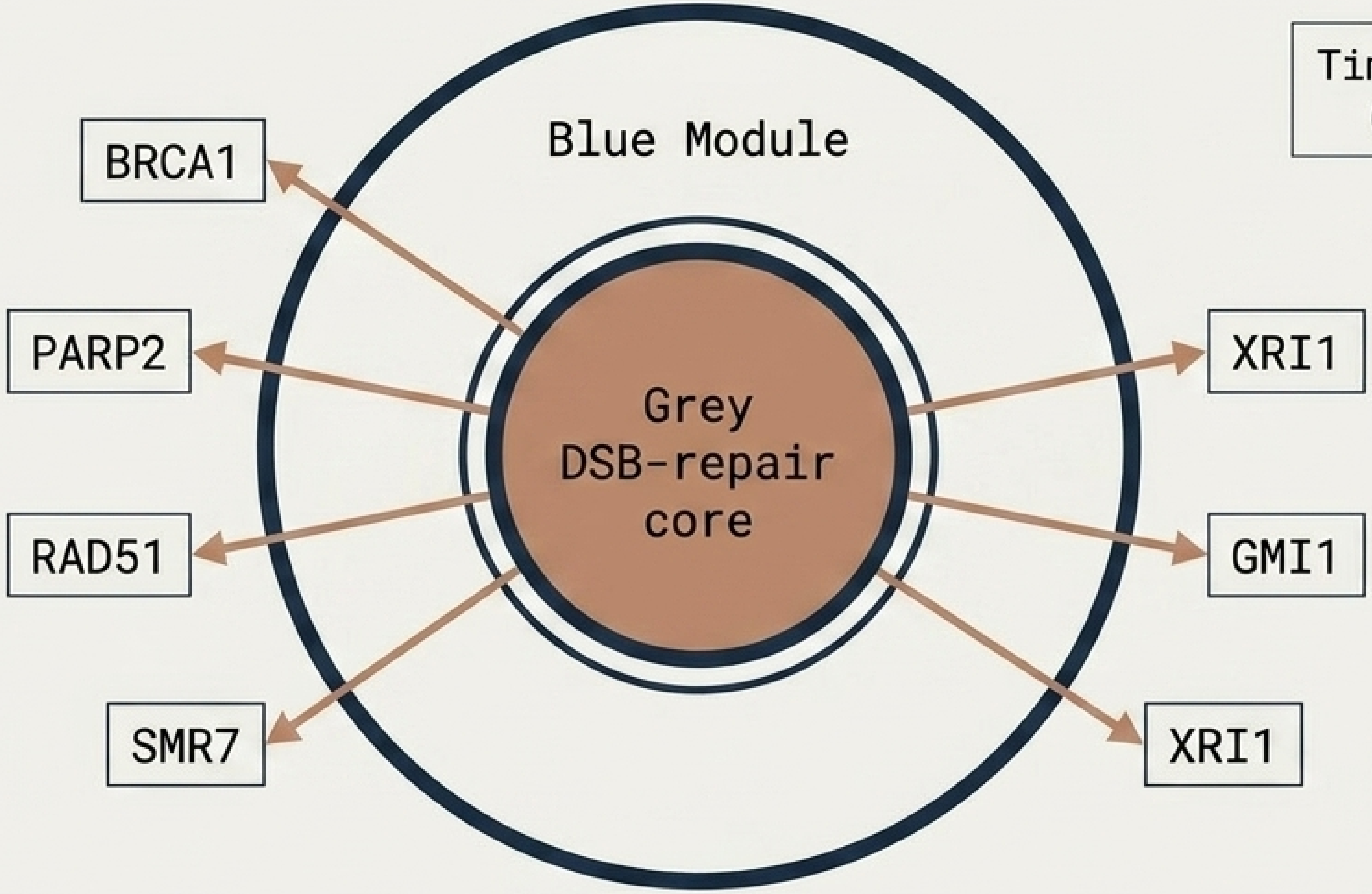
Timeframe: 0–2 hours.



The initial blue module is highly enriched for independently-called radiation Differentially Expressed Genes (DEGs), confirming immediate physiological detection.

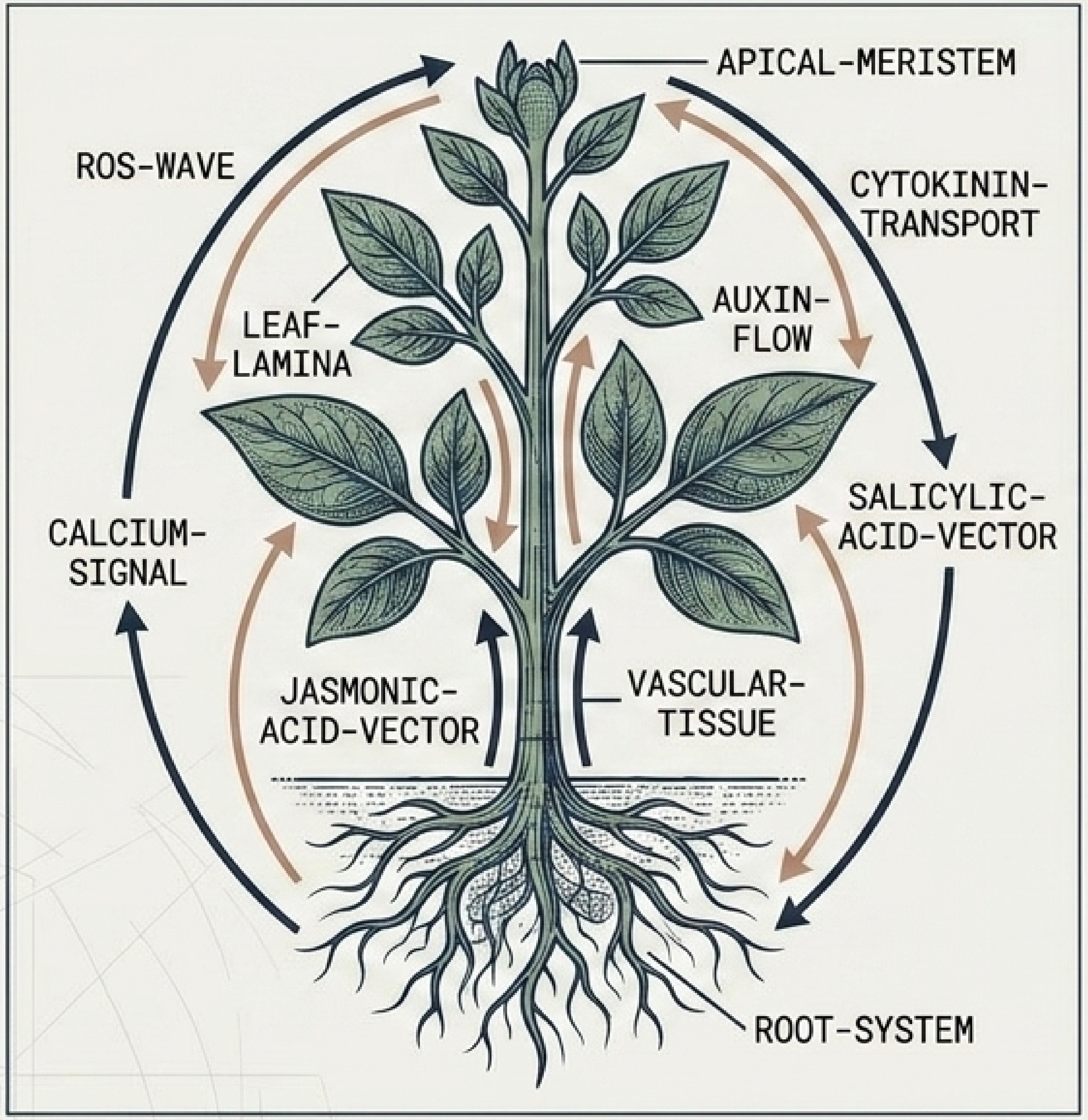
Phase 2: The Crisis Nadir and DSB-Repair Core Activation

Timeframe: 2–6 hours
(Maximum Stress)



At the precise moment the Resilience Index drops to 0.68, a targeted cluster of double-strand break (DSB) repair genes activates to stabilize the plant genome.

Phase 3: Spatial Recovery and Inter-Tissue Signaling



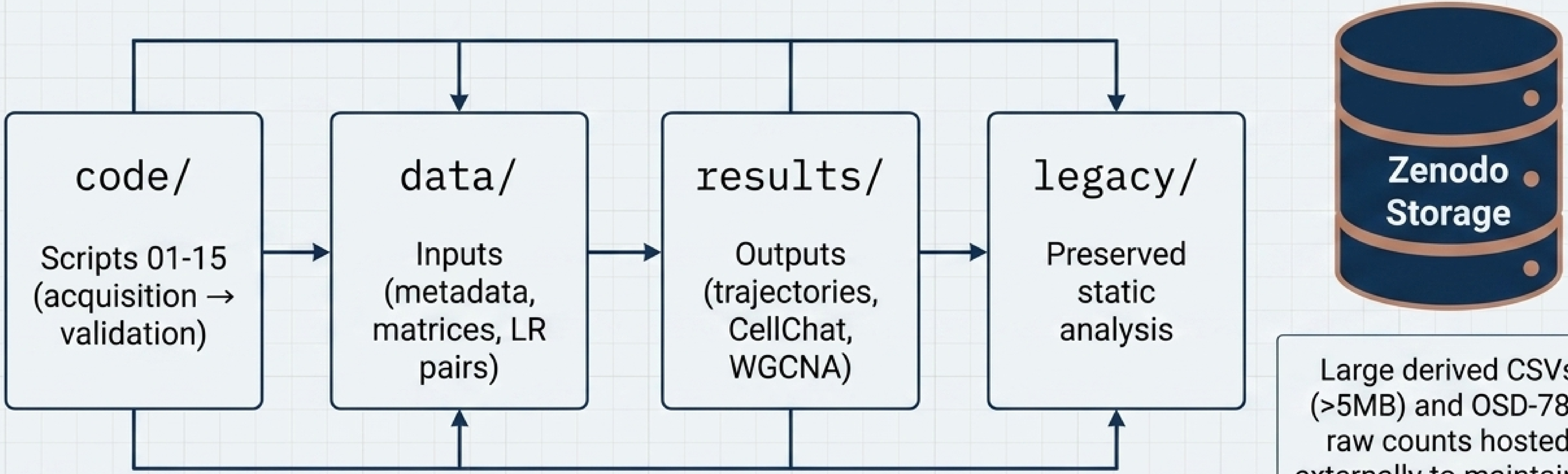
Timeframe: 6–24 hours
(Partial Recovery)

Post-crisis, stress-signaling waves propagate across the macro-structure of the plant (reconstructed via PlantCellChat) to re-establish biological homeostasis.

Independent Validation Confirms Model Predictive Power

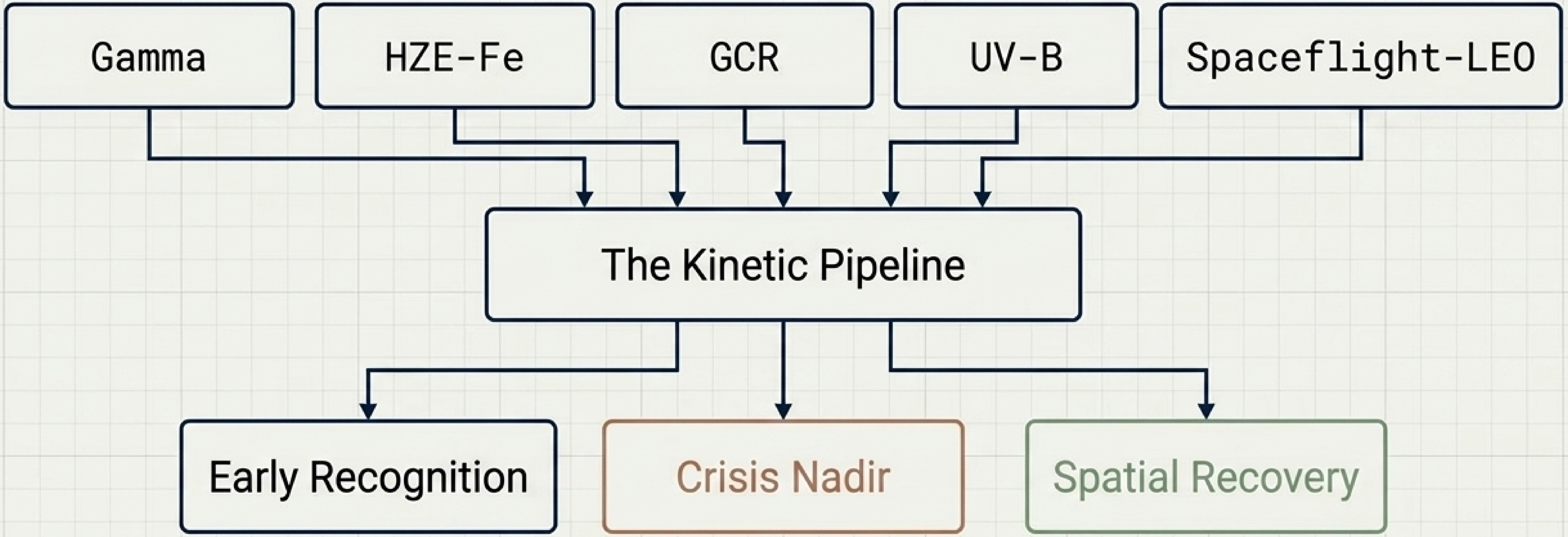
Training Cohort	Validation Cohort
10 Studies	**Study: OSD-782 / GLDS-679 (held-out)** **Condition: Low-dose gamma radiation** **Held-out response genes successfully recover the early-response blue module.** Enrichment: 2.3x | P-value: 2.8×10^{-12}

Repository Architecture & Reproducibility Blueprint



Large derived CSVs (>5MB) and OSD-782 raw counts hosted externally to maintain a lightweight main git repository.

From Raw Telemetry to Actionable Biological Targets



By harmonizing distinct OSDR datasets into a unified kinetic model, we isolate the precise spatial and temporal targets required for engineering radiation-resilient extraterrestrial agriculture.

Access, Licensing, and Source Data

Author: Dr. Richard Barker (@dr-richard-barker)

Repository: `github.com/dr-richard-barker/Plant_response_to_radiation`

Data License: CC-BY 4.0

Code License: MIT

Data sourced from the NASA Open Science Data Repository (osdr.nasa.gov)

NotebookLM